

# KELLY KAPSAR

Michigan State University - Department of Integrative Biology

 [github.com/kellykapsar](https://github.com/kellykapsar) |  [kelly.kapsar@gmail.com](mailto:kelly.kapsar@gmail.com) |  (314) 413-3271 |  [Kellykapsar.netlify.app](https://kellykapsar.netlify.app)

---

## EDUCATION

- 2022      **Ph.D. in Fisheries & Wildlife**  
Center for Systems Integration & Sustainability, Michigan State University  
University Distinguished Fellow  
*Advisor:* Dr. Jianguo (Jack) Liu  
*Thesis:* The metacoupled Arctic and north Pacific: Analyzing the spatiotemporal patterns and impacts of marine vessel traffic in coupled human and natural systems
- 2014      **B.A. in Biology, Certificate of Advanced Studies in Spanish**  
Carleton College, Northfield, MN  
Distinction in major, Magna Cum Laude  
*Thesis:* Who's afraid of the big bad wolf? Trait-mediated interactions among sympatric carnivores and their effect on behaviorally-mediated trophic cascades.

---

## RESEARCH INTERESTS

- Reproducible research
- Movement ecology
- Human-wildlife conflict
- Big data
- Remote sensing
- Spatial data analysis
- Open-source software
- Community engagement
- Spatial ecology
- Telecoupling
- Social-ecological systems
- Science communication

---

## PROFESSIONAL APPOINTMENTS

- 2023 – pres. **Data Scientist**  
Institute for Biodiversity, Ecology, Evolution, and Macrosystems,  
Integrative Biology Department, MSU  
Location: Remote  
Supervisor: Dr. Phoebe Zarnetske
- 2023 – pres. **Postdoctoral Research Associate**  
Spatial & Community Ecology Lab, Integrative Biology Department, MSU  
Location: Remote  
Supervisor: Dr. Phoebe Zarnetske
- 2022 – 2025 **Postdoctoral Research Associate**

Center for Systems Integration & Sustainability, Michigan State University  
Location: Remote  
Supervisor: Dr. Jianguo (Jack) Liu

---

## PEER-REVIEWED JOURNAL ARTICLES

Bills, P., Ramesh, A., Petri, L., Martínez-Blancas, A., **Kapsar, K.**, Tiwari, A., and Zarnetske, P. CollaboratoR: A scalable workflow for collaborative data entry and management. [*In review*].

**Kapsar, K.**, Kounta, L., Baiser, B., Bills, P., Record, S., Strecker, A. Zarnetske, P. (2026). Multi-scale environmental geodiversity: data for the National Ecological Observatory Network (NEON) with an adaptable workflow. *Scientific Data*, <https://doi.org/10.1038/s41597-026-07613-5>.

Waloven, S., Portales, N., **Kapsar, K.**, and Liu, J. (2025). A synthesized dataset of Pacific Arctic marine mammal occurrences (1860-2024). *Data in Brief*, e112036. <https://doi.org/10.1016/j.dib.2025.112036>

**Kapsar, K.**, Sullender, B., and Kuletz, K. (2025). A multi-scale seasonal analysis of vessel risk to seabirds in the North Pacific and Arctic. *Conservation Biology*, e70115. <https://doi-org/10.1111/cobi.70115>

Block, I., Remer, M., **Kapsar, K.**, and Schmidt, J. (2025). Analysis on the Effectiveness of International Maritime Organization's 'Areas to be Avoided' Policy on Vessel Traffic in the Bering Strait (2015-2022). *Marine Policy*, 181, 106849. <https://doi.org/10.1016/j.marpol.2025.106849>.

Youngflesh, C., **Kapsar, K.**, Uscanga, A., Williams, P.J., Doser, J.W., Kounta, L., and Zarnetske, P.L. (2024). Environmental variability shapes life history of the world's birds. *Ecology Letters*, 28 (2), e70077. <https://doi.org/10.1111/ele.70077>

**Kapsar, K.**, Gunn, G., Brigham, L., and Liu, J. (2023). Measuring recent increases in vessel traffic in the ice-covered waters of the Pacific Arctic. *Climatic Change*, 176 (7), 94. <https://doi.org/10.1007/s10584-023-03568-3>

Waloven, S., **Kapsar, K.**, Schwoerer, T., Berman, M., Schmidt, J., Vina, A., and Liu, J. (2023). Global gateways as telecoupled human and natural systems: The emerging case of the Bering Strait. *Ambio*. <https://doi.org/10.1007/s13280-023-01835-2>.

**Kapsar, K.**, Sullender, B., Liu, J., and Poe, A. (2022). North Pacific and Arctic marine traffic dataset (2015-2020). *Data in Brief*, 44, 108531. <https://doi.org/10.1016/j.dib.2022.108531>.

- Kapsar, K.**, Frans, V., Brigham, L., and Liu, J. (2022). The metacoupled Arctic: Human-nature interactions across local to global scales as drivers of sustainability. *Ambio* 51(10), 2061-2078. <https://doi.org/10.1007/s13280-022-01729-9>.
- Sullender, B., **Kapsar, K.**, Poe, A., and Robards, M. (2021). Spatial management measures alter vessel behavior in the Aleutian Archipelago. *Frontiers in Marine Science*, 7, 579905. <https://doi.org/10.3389/fmars.2020.579905>.
- Chung, M.G., **Kapsar, K.**, Frank, K., and Liu, J. (2020). The spatial and temporal dynamics of global meat trade networks. *Scientific Reports*, 10(1), 16657. <https://doi.org/10.1038/s41598-020-73591-2>.
- Kapsar, K.**, Hovis, C., Bicudo da Silva, R., Buchholtz, E., Carlson, A., Dou, Y., ... Liu, J. (2019). Telecoupling Research: The First Five Years. *Sustainability*, 11(4), 1033. <https://doi.org/10.3390/su11041033>
- Herzberger, A., Chung, M. G., **Kapsar, K.**, Frank, K. A., and Liu, J. (2019). Telecoupled Food Trade Affects Pericoupled Trade and Intracoupled Production. *Sustainability* 11(10), 2908. <https://doi.org/10.3390/su11102908>.
- Xu, Z., Li, Y., Herzberger, A., Chen, X., Gong, M., **Kapsar, K.**, ... & Liu, J. (2019). Interactive national virtual water-energy nexus networks. *Science of The Total Environment*, 673: 128-135. <https://doi.org/10.1016/j.scitotenv.2019.03.298>.
- Du, Y., Wan, Q., Liu, H., **Kapsar, K.**, Peng, J. (2019). How does urbanization influence PM2.5 concentrations? Perspective of spillover effect of multi-dimensional urbanization impact. *Journal of Cleaner Production*, 220: 974-983. <https://doi.org/10.1016/j.jclepro.2019.02.222>

---

## PEER-REVIEWED BOOK CHAPTERS & REPORTS

- Beck, J. M., Hoffmann, C. F., **Kapsar, K.**, Montgomery, R. A. (2026). The power of value-laden framing: examples from wildlife conservation research. In *Routledge Handbook of Values and Science*. <https://doi.org/10.4324/9781003469100-49>.
- Sullender, B., **Kapsar, K.**, and Kuletz, K. (2022). Appendix 6: Seabird-vessel traffic risk analysis for Alaska's marine regions. In *Marine Bird Distribution and Abundance in Offshore Waters*. BOEM 2022-067. <https://www.boem.gov/regions/alaska-ocs-region/scientific-and-technical-publications-2022>
- Berkman, P., Fiske, G., Loenzini, D., Young, O., Pletnikoff, K., Grebmeier, J., Fernandez, L., Devinie, L., Causey, D., **Kapsar, K.**, and Jørgensen, L. (2022). Satellite Record of Pan-Arctic Maritime Ship Traffic. In *Arctic Report Card 2022*, M. L. Druckenmiller, R. L. Thoman, and T. A. Moon, Eds. <https://doi.org/10.25923/mhrv-gr76>.

- Liu, J., Herzberger, A, **Kapsar, K.**, Carlson, A., and Connor, T. (2019). “What is telecoupling?”. In J. Østergaard Nielsen and C. Friis (Eds.), *Telecoupling: Exploring Land Use Change in a Globalized World* (pp. 19-48). London: Palgrave MacMillan. <https://doi.org/10.1007/978-3-030-11105-2>.
- Cerezo, A., & **Kapsar, K. E.** (2018). Zoo Conservation Disembarks: Stepping off the Ark and into Global Sustainable Development. In B. A. Minter, J. Maienschein, & J. P. Collins (Eds.), *The Ark and Beyond: The Evolution of Zoo & Aquarium Conservation* (pp. 344–359). Chicago: University of Chicago Press. <https://doi.org/10.1007/978-3-030-11105-2>.

---

## DATA SETS

- Kapsar, K.**, Kounta, L., Bills, P., Record, S., Strecker, A., Baiser, B., Zarnetske, P. (2025). Multi-scale environmental geodiversity: data for the National Ecological Observatory Network (NEON) with an adaptable workflow ver 2. Environmental Data Initiative.
- Waloven, S., Portales, N., **Kapsar, K.**, and Liu, J. (2025). A synthesis of Pacific Arctic marine mammal occurrences (1860-2024). Arctic Data Center. <https://doi.org/10.18739/A2WH2DH0V>
- Youngflesh, C., **Kapsar, K.**, Uscanga, A., Williams, P.J., Doser, J.W., Kounta, L., and Zarnetske, P.L. (2024). Inter- and intra-annual temperature and precipitation variability (1950-2022) across the ranges of non-migratory birds and their association with generation length ver 1. Environmental Data Initiative. <https://doi.org/10.6073/pasta/59fba1b7dfe96856b17143f8e50f5722>
- Kapsar, K.**, Sullender, B. & Poe, A. (2022). North Pacific and Arctic Marine Vessel Traffic Dataset (2015-2020); 25 Kilometer Resolution. Arctic Data Center. doi:10.18739/A2J678Z16.

---

## GRANTS, FELLOWSHIPS, & AWARDS (Total: \$2,024,950)

- |      |                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Environmental Defense Fund (Solar Radiation Modification) (\$270,000)                                                                                                         |
| 2025 | MSU Postdoctoral Association Travel Funding (\$400)                                                                                                                           |
| 2022 | MSU Institute for Biodiversity, Ecology, Evolution, and Macrosystems (IBEEM) Postdoctoral Fellowship (\$75,000)                                                               |
| 2021 | College of Agriculture and Natural Resources Dissertation Completion Fellowship (\$10,000)                                                                                    |
| 2021 | NSF Arctic System Science. <i>Collaborative Research: Complex Effects of Telecoupled Activities in the Changing Environment of the Arctic</i> (Award #2033507) (\$1,541,800)* |
| 2021 | College of Agriculture and Natural Resources Summer Fellowship (\$7,000)                                                                                                      |

|      |                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------|
| 2021 | College of Agriculture and Natural Resources Alumni Association Scholarship (\$3,000)                                           |
| 2021 | Ambrose Patullo Graduate Fellowship in Literary Works (\$2,500)                                                                 |
| 2020 | American Geophysical Union (AGU) Virtual Student Travel Grant (\$1,000)                                                         |
| 2019 | Gilbert W. Mouser Scholarship in Conservation & Environmental Education (\$2,500)                                               |
| 2018 | William W. and Evelyn M. Taylor Endowed Fellowship for International Engagement in Coupled Human and Natural Systems (\$28,000) |
| 2018 | MSU Graduate School Travel Funding (\$1,050)                                                                                    |
| 2018 | MSU Fisheries and Wildlife Graduate Student Organization Travel Grant (\$200)                                                   |
| 2017 | Vera M. Wallach Graduate Fellowship (\$2,500)                                                                                   |
| 2016 | University Distinguished Fellowship, Michigan State University (\$80,000)                                                       |

---

\*Principal contributor to Project Description and Data Management Plan.

---

## TEACHING

### Courses

|             |                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025        | <b>Instructor</b><br>Spatial Ecology (IBIO870) – Fall 2025<br>Integrative Biology Department, Michigan State University                                      |
| 2020        | <b>Instructor</b><br>Organismal and Population Biology Laboratory (BS172L)<br>Biological Sciences Department, Michigan State University                      |
| 2019        | <b>Instructor</b><br>Applications in Biological Science Laboratory (ISB208L)<br>Center for Integrative Studies in General Science, Michigan State University |
| 2013 - 2014 | <b>Teaching Assistant</b><br>Population Ecology (BIOL352); Climate Change Geology (GEOL115)<br>Carleton College                                              |

### Workshops & Guest Lectures

|      |                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>Guest Lecture</b><br>“Open Science: Strengthening Trust and Reproducibility”<br>Integrative Biology Department, Michigan State University |
| 2024 | <b>Co-Instructor</b><br>“Collaborative Coding for Scientific Research: Git, GitHub, and open science”                                        |

---

## RESEARCH, OUTREACH, & FIELD EXPERIENCES

- 2015 – 2025 **Research Associate, Community Engagement in Conservation**  
Saint Louis Zoo Arctic Socio-Environmental WildCare Program, St. Louis, MO
- 2023 **Data Scientist**  
ACTNOW Amazonas Action Alliance, XPRIZE Rainforest, East Lansing, MI
- 2019 **Research Assistant**  
Healy Research Cruise 1901, Bering & Chukchi Seas, AK
- 2018 – 2019 **Social Media Associate**  
Engagement Scholarship Consortium, East Lansing, MI
- 2014 – 2016 **Outreach Instructor**  
Saint Louis Zoo Education Department, Saint Louis, MO
- 2014 **Conservation Education Research Extern**  
Saint Louis Zoo Education Department, Saint Louis, MO

---

## CONFERENCES & INVITED PRESENTATIONS

- 2026 **ESA Open Science Section Webinar Series**  
Virtual  
*Presentation:* AI in Ecological Research: Approaches to and applications of open access models
- 2026 **Alaska Marine Science Symposium**  
Anchorage, Alaska  
*Presentation:* Navigating change: Mapping vessel dynamics in the rapidly evolving Bering Strait
- 2025 **University of Illinois at Chicago**  
Chicago, Illinois  
*Invited Seminar:* Multi-scale environmental geodiversity and climate data layers for the National Ecological Observatory Network (NEON) analyses across scales
- 2025 **Ecological Society of America**  
Baltimore, Maryland

*Presentation:* Multi-scale environmental geodiversity and climate data layers for the National Ecological Observatory Network (NEON) analyses across scales

- 2025      **International Association of Landscape Ecology (N. America Chap.)**  
Raleigh, North Carolina  
*Presentation:* Navigating change: Mapping metacoupled vessel dynamics in the rapidly evolving Bering Strait
- 2024      **International Association of Landscape Ecology (N. America Chap.)**  
Oklahoma City, Oklahoma  
*Poster:* Recent patterns of vessel risk to seabirds: A multi-scale seasonal analysis in the North Pacific and Arctic.
- 2024      **NSF Macrosystems Biology Meeting**  
Virtual  
*Poster:* Data and Workflows for Macrosystems Biodiversity: Climate, Geodiversity, and Disturbance.
- 2024      **International Sea Duck Conference**  
Virtual  
*Presentation:* Quantifying risk of vessel encounters with sea ducks in the North Pacific: A multi-scale seasonal analysis of Alaskan Oceans.
- 2023      **Alaska Marine Science Symposium**  
Anchorage, Alaska  
*Presentation:* Seabird-vessel traffic risk analysis for Alaska's oceans
- 2022      **American Geophysical Union**  
Chicago, Illinois  
*Presentation:* The Metacoupled Arctic: Human–Nature Interactions Across Local to Global Scales as Drivers of Sustainability
- 2022      **Northern Latitudes Partnerships-USFWS Monthly Outreach Meeting**  
Virtual  
*Presentation:* North Pacific and Arctic Marine Vessel Traffic Data Set (2015-2020)
- 2022      **Alaska Marine Policy Forum**  
Virtual  
*Presentation:* Quantifying the spatio-temporal dynamics of vessel traffic in a rapidly changing maritime bottleneck
- 2022      **USFWS Oil Spill Team Meeting**  
Virtual

*Presentation:* A new, publicly available marine vessel traffic data set for the north Pacific and southern Arctic Oceans, 2015-2020

2022 **Northern Latitudes Partnership**

Virtual

*Presentation:* A new, publicly available marine vessel traffic data set for the north Pacific and southern Arctic Oceans, 2015-2020

2022 **Alaska Marine Science Symposium**

Virtual

*Presentation:* A new, publicly available marine vessel traffic data set for the north Pacific and southern Arctic Oceans, 2015-2020

2021 **Arctic Vessel Traffic Working Group**

Virtual

*Presentation:* Building a public vessel traffic data set  
(Co-presented)

2021 **Graduate Women in Science Research Seminar**

Virtual

*Presentation:* Fish and ships: Evaluating the influence of commercial fishing on Steller sea lion habitat selection

2020 **5<sup>th</sup> Distributed Biological Observatory (DBO) Data Meeting**

Seattle, Washington

*Presentation:* Linking coupled human and natural systems approaches with the Distributed Biological Observatory

2018 **US Chapter of the International Association of Landscape Ecology**

Chicago, Illinois

Telecoupling symposium organizer

*Presentation:* Quantifying the spatio-temporal distribution of marine vessel traffic in the Bering Strait

2017 **CN Forum: Putting the Arctic in Plain Sight**

Lansing, Michigan

*Poster:* The Telecoupled Arctic: Marine Vessel Traffic in the Bering Strait Region

2017 **Smithsonian National Museum of the American Indian**

Washington, D.C.

*Presentation:* Zebras in the Arctic: Branching out from the McDonnell Polar Bear Point Project (Recording:

[https://www.youtube.com/watch?](https://www.youtube.com/watch?v=YrOVJQPDNEQ&ab_channel=SmithsonianNMAI)

[v=YrOVJQPDNEQ&ab\\_channel=SmithsonianNMAI](https://www.youtube.com/watch?v=YrOVJQPDNEQ&ab_channel=SmithsonianNMAI))

---

## INVITED PANELS



- 2018      **Women in Nature Network**  
Lake Mainstay, Guyana  
Invited Panelist
- 2017      **MSU Summer Intensive on Community Engagement**  
Lansing, Michigan  
Invited Panelist

---

## JOURNAL AFFILIATIONS

Reviewer: Ambio  
Reviewer: Ecology & Society  
Reviewer: Journal of Cleaner Production

---

## PROFESSIONAL PUBLICATIONS

Block, I., Waloven, S., Remer, M., Akogun, R., and **Kapsar, K.** (2023). Shipping Across the Arctic: Take a deep dive into the Bering Strait to explore the dynamic relationships between ships, communities, and wildlife.  
<https://storymaps.arcgis.com/stories/8b3d286071cf4df2a443c0ccfbecda60>

**Kapsar, K.** (2021). Breaking the Ice: Connecting data and reality on an Arctic research cruise. Spotlight Magazine. <https://spotlight-msu.medium.com/breaking-the-ice-connecting-data-and-reality-on-an-arctic-research-cruise-a8f8028ff7f4>

**Kapsar, K.** (2018). Zoos should leave the ark metaphor behind. Blog post for Nature Ecology and Evolution blog.  
<https://natureecoevocommunity.nature.com/users/178448-kelly-kapsar/posts/39133-zoos-should-embrace-the-un-sustainable-development-goals-to-achieve-their-wildlife-conservation-missions>

**Kapsar, K.**, McCord, P, and Tonini, F. (2018). The Bering Sea International Commodity Transport. Case study for Telecoupling GeoApp. <https://telecoupling.msu.edu/geo-app/commodity-transport/>

---

## DEPARTMENTAL & PROFESSIONAL SERVICE

- 2025-pres.    **International Association for Landscape Ecology – N. America Chapter**  
- Awards Committee Member
- 2022 – 2024   **Fisheries & Wildlife Department Diversity, Equity, & Inclusion Committee**, Michigan State University  
- Postdoc Representative

- 2018 – 2021 **Women in Nature Network – MSU Chapter**
- Founding Member
  - Social Media Associate (2019-2020)
  - Secretary (2018-2019)
- 2020 **Outreach & Engagement Reopening Subcommittee**, Michigan State University
- Graduate Student Representative
- 2017 – 2020 **Fisheries & Wildlife Department Graduate Student Organization**, Michigan State University
- Symposium Co-Chair (2019-2020)
  - Symposium Committee (2017-2019)
  - Spotlight Magazine Editor (2017-2020)
  - Curriculum Committee (2017-2018)
  - Outreach Committee (2017-2018)
- 2018 – 2020 **Wildside Wildlife Rehabilitation Center**
- Volunteer Wildlife Rehabilitator
  - Environmental Educator
- 2019 – 2020 **Association of Polar Early Career Scientists (APECS)**
- Council Member

---

## PROFESSIONAL SKILLS & MEMBERSHIPS

|                |                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Software       | R, Rspatial, Python, Git, GitHub, ArcGIS, Erdas Imagine, Stata, Microsoft Office Suite (incl. Excel, Access, PowerPoint, Word)                                                              |
| Languages      | English (fluent), Spanish (proficient)                                                                                                                                                      |
| Visual media   | Basic videography and graphics generation skills (R, PowerDirector, Adobe Photoshop)                                                                                                        |
| Memberships    | Ecological Society of America,<br>Women in Nature Network,<br>International Association for Landscape Ecology – N. America chapter                                                          |
| Certifications | NSF Arctic Data Center: Scalable and Computationally Reproducible Approaches to Arctic Research (2023)<br>NSF Arctic Data Center: Reproducible Practices for Arctic Research Using R (2019) |

SCUBA (PADI Open Water)  
Ready for Media Training (2016)